

# Graph-Aware Anomaly Detection for Blockchain Forensics: A 48-Experiment Benchmark on the Elliptic Bitcoin Dataset

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**Abstract**—Detecting illicit activity on a public blockchain involves facing three concurrent challenges: severe class imbalance (2 % illicit among all transactions), temporal concept drift (illicit rate shifting from 11.6 % in training to 6.5 % at test time), and a pseudonymous graph whose nodes carry 165 anonymous features. We present a systematic benchmark of 48 model variants across five paradigms—unsupervised, supervised, deep neural, graph neural network (GNN), and semi-supervised—on the Elliptic Bitcoin dataset (203 769 transactions, 234 355 directed edges), organized into five iterative improvement rounds. All models share a strict temporal train/test split that respects the time-series nature of the data. The best result,  $F1 = 0.8265$ , is achieved by Gradient Boosting augmented with seven structural topology features. GraphSAGEv2 reaches  $F1 = 0.698$  using graph message passing over node features and topology, recovering 84.5 % of the supervised ceiling. Classical feature-space unsupervised baselines never exceed  $F1 = 0.076$ ; the course-derived DOMINANT graph autoencoder reaches  $F1 = 0.212$ . A controlled GNN ablation shows that LayerNorm and self-loops (+0.068  $F1$ ) matter more than depth or attention.  $\beta$ -VAE with  $\beta=0.01$  achieves  $F1 = 0.407$ , the best purely unsupervised result.

**Index Terms**—anomaly detection, blockchain forensics, GraphSAGE, DOMINANT, gradient boosting,  $\beta$ -VAE, EVT threshold, SHAP, Optuna, class imbalance, Elliptic dataset

## I. INTRODUCTION

Bitcoin’s blockchain is fully public yet pseudonymous: every transaction is permanently recorded, but the identities behind wallet addresses remain hidden. This pseudonymity has made Bitcoin a vehicle for money laundering, ransomware payments, and darknet market settlements. Traditional anti-money-laundering (AML) controls rely on verified identity, while blockchain forensics must infer intent from transaction patterns and graph structure.

The Elliptic dataset [1] is the largest publicly available labeled graph of real Bitcoin transactions: 203 769 transactions, 234 355 directed edges, 4 545 labeled illicit (2 %), 42 019 labeled licit (21 %), and 157 205 unlabeled (77 %) over 49 time steps spanning roughly two years. Each node carries 165 anonymous features, 93 local statistics and 71 aggregated neighborhood summaries (some sources count 166 including the time-step index, which we treat as a split attribute rather than a predictive feature).

## Three intertwined challenges:

- 1) **Severe imbalance.** 9.76 % illicit among labeled nodes; 6.5 % in the test window. A naive majority classifier achieves 93.5 % accuracy while detecting zero illicit transactions, making accuracy a completely uninformative metric.
- 2) **Temporal concept drift.** Illicit rate drops from 11.6 % to 6.5 % from train to test (44 % relative decline). Random shuffling leaks future patterns into training, inflating every metric.
- 3) **Anonymous features.** Elliptic has not published feature semantics, so cross-model feature agreement is the only validation of signal robustness.

## A. Research Questions

- **RQ1:** Do classical unsupervised detectors provide a viable AML baseline, or do illicit transactions fail to form anomalous clusters?
- **RQ2:** How much does graph structure add over tabular supervised models, and what fraction of the supervised ceiling can a GNN recover?
- **RQ3:** Which GNN design choices (depth, width, normalization, attention) matter most on a sparse graph (avg. degree 2.3)?
- **RQ4:** Do any features rank consistently across heterogeneous model families?

## B. Contributions

- 1) Strict **temporal evaluation protocol** (steps 1–34 train / 30–34 validation / 35–49 test) applied to all 48 variants.
- 2) **48-variant benchmark** across five improvement rounds covering classical anomaly detection, supervised trees, deep autoencoders, GNNs, semi-supervised pseudo-labels, and topology-augmented methods.
- 3) **10-experiment GNN ablation** isolating hidden dimension, depth, LayerNorm, self-loops, and attention heads.
- 4) **Optuna TPE tuning** [14] (40 trials each) for GBM, RF, and GraphSAGE.

- 5) Evaluation of **course-derived methods**: DOMINANT [6], LSTM-AE,  $\beta$ -VAE [7], spectral embedding, EVT thresholding [15], and pseudo-label propagation [16].
- 6) **Cross-paradigm explainability** via SHAP and gradient attribution.

## II. RELATED WORK

**Weber et al.** [1] introduced the Elliptic dataset and benchmarked LR, RF, and GCN, but did not explore unsupervised methods, deeper GNN architectures, or the score-inversion phenomenon in reconstruction-based detectors.

**GNNs for fraud.** Hamilton et al. [3] introduced GraphSAGE; Velickovic et al. [4] proposed GAT; Brody et al. [5] fixed GAT’s expressivity with GATv2. Pareja et al. [2] extended GCN to dynamic graphs (EvolveGCN).

**Graph anomaly detection.** Ding et al. [6] proposed DOMINANT, a GCN autoencoder that jointly reconstructs node attributes and graph structure without labels.

**Reconstruction-based detection.** Standard autoencoders [8] flag high reconstruction error as anomalous. On Elliptic we observe inversion: illicit transactions reconstruct with *lower* MSE [9]. Denoising AE [10] and VAE [11] are evaluated.

$\beta$ -VAE [7] adds weight  $\beta$  to the KL term in the ELBO:

$$\mathcal{L}_\beta = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - \beta D_{\text{KL}}(q_\phi(z|x) \| p(z)). \quad (1)$$

Lower  $\beta$  is reconstruction-dominated;  $\beta=0.01$  yields best F1 (0.407) since reconstruction dominance preserves the inversion signal.

**EVT thresholding.** The Generalized Pareto Distribution (GPD) models score tails to set thresholds without labels [15]:

$$t_{\text{EVT}} = u + \frac{\hat{\sigma}}{\hat{\xi}} \left( \left( \frac{np}{N_u} \right)^{-\hat{\xi}} - 1 \right), \quad (2)$$

where  $u$  is the tail cutoff,  $N_u$  exceedances, and  $p$  exceedance probability.

## III. DATASET AND METHODOLOGY

### A. Dataset

TABLE I: Elliptic Dataset Statistics

| Property          | Value   | Note                  |
|-------------------|---------|-----------------------|
| Transactions      | 203 769 | total nodes           |
| Directed edges    | 234 355 | avg. degree 2.30      |
| Illicit (label 1) | 4 545   | 2 % of total          |
| Licit (label 2)   | 42 019  | 21 % of total         |
| Unlabeled         | 157 205 | 77 % of total         |
| Time steps        | 49      | steps 44–49 unlabeled |
| Local features    | 93      | 1f_1–1f_93            |
| Aggregated feat.  | 71      | af_1–af_71            |

The Elliptic dataset was published by Elliptic in collaboration with MIT. Feature semantics are not published; columns carry only numerical indices.

TABLE II: Temporal Data Split (by time step index, 1–49)

| Partition                 | Labeled rows     | Illicit % |
|---------------------------|------------------|-----------|
| Train (steps 1–34)        | 29 894           | 11.6 %    |
| Validation (steps 30–34)  | subset of train  | —         |
| Test (steps 35–49)        | 16 670           | 6.5 %     |
| Unsup. train (steps 1–34) | 136 265 all rows | —         |
| Licit-only (steps 1–34)   | 26 432           | 0 %       |

### B. Temporal Split and Evaluation Protocol

A strict temporal split is mandatory: Bitcoin transactions have causal ordering, and random shuffling inflates all metrics by allowing interpolation between past and future behavior.

The 44 % relative drop in illicit rate confirms genuine concept drift. **Primary metric**: F1 on the illicit class. Secondary: PR-AUC (threshold-independent, informative under imbalance) and ROC-AUC.

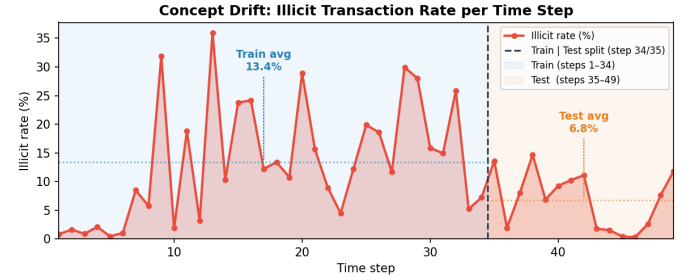


Fig. 1: Illicit transaction rate per time step (labeled nodes only). The rate falls from roughly 11.6 % in the training window (steps 1–34) to 6.5 % in the test window (steps 35–49), a 44 % relative decline confirming genuine concept drift. Random shuffling would mask this shift and inflate all metrics.

### C. Per-Paradigm Training Setup

Each paradigm uses different training data and threshold strategy, as summarised in Table III. *Note on supervision*: models in the IsoForest/LOF and OCSVM/AE/VAE rows use no labels for *training*, but their classification thresholds are selected by optimizing F1 on a labeled subset. They are therefore better described as *semi-supervised at threshold-selection stage* rather than fully unsupervised; truly label-free thresholds (EVT) are evaluated separately.

TABLE III: Per-Paradigm Training and Threshold Strategy

| Paradigm         | Training data           | Threshold               |
|------------------|-------------------------|-------------------------|
| IsoForest / LOF  | 136 265 all train rows  | F1-opt, labeled train   |
| OCSVM / AE / VAE | 26 432 licit-only rows  | F1-opt, val steps 30–34 |
| Supervised       | 29 894 labeled train    | 0.5 (default)           |
| GNN              | Full graph, train masks | 0.5 (default)           |
| Ensemble         | Labeled train rows      | Grid search, val steps  |
| Semi-sup.        | Train + pseudo-labeled  | 0.5 (default)           |

#### D. Model Families

1) *Classical Unsupervised*: Isolation Forest [12] (contamination=0.1); LOF (transductive, applied to test set directly); One-Class SVM (RBF,  $\nu=0.05$ , licit-only train). Also evaluated: IsoForest on 7 structural topology features, and IsoForest/OCSVM on the top-50 spectral eigenvectors of  $\tilde{A} = D^{-1/2}AD^{-1/2}$ .

2) *Supervised Tree Models*: All use `class_weight='balanced'` or `scale_pos_weight`: Logistic Regression (L2), Random Forest (100 trees), Gradient Boosting (Optuna-tuned:  $n=245$ ,  $\text{depth}=8$ ,  $\text{lr}=0.121$ ,  $\text{sub}=0.918$ ), LightGBM (500 trees,  $\text{max\_depth}=8$ ), RF + graph degree/PageRank features, GBM + 7 structural topology features (degree, clustering, ego-density, OddBall), GBM + 330 temporal rolling features (5-step window), and GBM retrained on pseudo-labeled unlabeled nodes.

3) *Neural Anomaly Detection*: All trained on licit-only data; anomaly score =  $-\text{MSE}$  (direction auto-detected on validation via AUC comparison): AE (166 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ ... $\rightarrow$ 166), DenoisingAE (latent=8,  $\sigma_{\text{noise}}=0.1$ ), VAE and  $\beta$ -VAE grid search ( $\beta \in \{0.01, 1, 2, 4, 8\}$ ) [7], [11], LSTM-AE (per-timestep mean features,  $T=10$  sliding windows), and DOMINANT (2-layer GCN encoder, joint attribute+structure reconstruction) [6].

4) *GNN (Supervised)*: Full PyG graph; edges made undirected; inverse-frequency `CrossEntropyLoss`. GraphSAGE mean aggregation at layer  $k$ :

$$h_v^{(k)} = \sigma \left( W^{(k)} \cdot \text{MEAN}(\{h_u^{(k-1)}\} \cup \{h_u^{(k-1)} : u \in \mathcal{N}(v)\}) \right). \quad (3)$$

Variants: baseline (2L,  $h=128$ , 200ep), GraphSAGEv2 (3L,  $h=256$ , LayerNorm, self-loops, 250ep), Optuna-tuned ( $\text{lr}=0.0038$ ,  $\text{dropout}=0.45$ , 274ep), SAGEv2 + pseudo-labels (130 655 nodes), GAT (2L, 4 heads), and GATv2 (3L, 2 heads, residual, LayerNorm).

5) *Ensemble Methods*: Soft-vote ensemble of RF+LGB+GAT (threshold on train, Round 1) and GBM+LGBM+SAGEv2 (threshold grid-searched on validation). Four weight schemes tested; equal weights (1/3 each) won.

#### E. EVT Thresholding

GPD fitted to the upper tail of reconstruction scores (tail quantile  $q=0.90$ , exceedance probability  $p=0.065$ ) per Eq. 2. Evaluated on all AE-based models.

#### F. Implementation

NVIDIA RTX 5070 Ti (12 GB VRAM, CUDA 12.8); Python 3.12, PyTorch 2.11, PyG 2.8, scikit-learn 1.9, LightGBM 4.6, SHAP 0.52, Optuna 4.9, NetworkX 3.6. `StandardScaler` fitted on train only; random seed 42 throughout.

### IV. EXPERIMENTS

Experiments follow a logical progression: (1) establish unsupervised floor (RQ1); (2) quantify supervised ceiling and tuning gains; (3) diagnose neural reconstruction ceiling; (4) systematic GNN ablation (RQ3); (5) course-derived advanced

methods; (6) cross-paradigm explainability (RQ4). All models are evaluated on the same 16 670 labeled test rows with no test-set leakage.

#### A. Unsupervised Baselines

TABLE IV: Classical Unsupervised Anomaly Detection

| Model                     | F1    | ROC-AUC | PR-AUC |
|---------------------------|-------|---------|--------|
| Isolation Forest          | 0.021 | 0.172   | 0.036  |
| One-Class SVM             | 0.023 | 0.235   | 0.039  |
| LOF                       | 0.076 | 0.506   | 0.066  |
| IsoForest (structural)    | 0.094 | 0.458   | 0.056  |
| IsoForest (spectral K=50) | 0.056 | 0.407   | 0.052  |
| OCSVM (spectral K=50)     | 0.061 | 0.391   | 0.054  |

All detectors score below  $F1 = 0.10$ . LOF’s AUC of 0.506 is random. Spectral embedding fails because the Fiedler value  $\approx 2.0$  (near bipartite maximum) indicates no sparse cut separating illicit nodes—money laundering chains are embedded within the spectral normal subspace.

**Answer to RQ1:** illicit transactions do *not* form anomalous clusters in feature space. Automated laundering produces stereotyped, regular transactions indistinguishable from high-frequency legitimate activity by density-based methods.

#### B. Supervised Models

TABLE V: Supervised Model Results (sorted by F1)

| Model                   | F1           | ROC-AUC | PR-AUC |
|-------------------------|--------------|---------|--------|
| Logistic Regression     | 0.302        | 0.881   | 0.291  |
| GBM (baseline)          | 0.766        | 0.914   | 0.785  |
| RandomForest (baseline) | 0.801        | 0.935   | 0.794  |
| RF + Graph Features     | 0.807        | 0.942   | 0.800  |
| LightGBM                | 0.817        | 0.930   | 0.804  |
| GBM + Temporal Rolling  | 0.818        | 0.930   | 0.802  |
| GBM (Optuna tuned)      | 0.824        | 0.920   | —      |
| GBM + PseudoLabels      | 0.824        | 0.924   | 0.804  |
| <b>GBM + Structural</b> | <b>0.827</b> | 0.924   | 0.803  |

Logistic Regression achieves recall = 0.87 but precision = 0.18 ( $F1 = 0.302$ )—unusable without heavy post-filtering. The jump from LR to RF ( $\Delta = +0.499$ ) reflects non-linear feature interactions that linear models cannot capture. Adding graph-topology features to RF gives only  $\Delta = +0.006$  because the 71 aggregated features already encode neighborhood information in tabular form.

GBM Optuna reaches  $F1 = 0.824$  ( $\Delta = +0.058$  over baseline): the learning-rate–depth–subsample interaction is non-obvious and benefits from systematic search. RF Optuna ( $\Delta = -0.004$ ) confirms RF is near its ceiling.

GBM with 7 structural topology features reaches  **$F1 = 0.8265$** , the overall best result ( $\Delta = +0.002$  over Optuna GBM): topology information is not fully subsumed by the 71 aggregated features.

GBM with 330 temporal rolling features (5-step window) improves AUC (0.920  $\rightarrow$  0.930) but reduces F1 (0.824  $\rightarrow$

0.818): global temporal context adds noise to per-transaction discrimination.

Semi-supervised GBM with pseudo-labels reaches  $F1 = 0.8235$  ( $\Delta = -0.0006$ ): GBM has already learned the optimal boundary from 29 894 labeled rows.

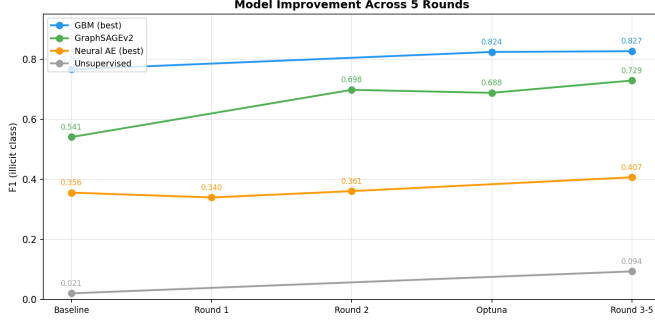


Fig. 2: Improvement across five rounds for key model families. “GBM (best)” tracks the highest-scoring GBM variant at each stage; “Neural AE (best)” tracks the highest-scoring autoencoder variant. Both labels include “(best)” to indicate they represent the family ceiling, not a single fixed model. GBM peaks at  $F1 = 0.827$ ; GraphSAGEv2 reaches 0.729; the autoencoder family peaks at 0.407 ( $\beta$ -VAE,  $\beta=0.01$ ).

### C. Neural Anomaly Detection

TABLE VI: Neural Anomaly Detection Results (sorted by F1)

| Model                         | F1           | ROC-AUC | PR-AUC |
|-------------------------------|--------------|---------|--------|
| LSTM-AE                       | 0.122        | 0.446   | 0.064  |
| VAEv2                         | 0.256        | 0.732   | 0.161  |
| $\beta$ -VAE ( $\beta=2.0$ )  | 0.280        | 0.746   | 0.211  |
| $\beta$ -VAE ( $\beta=4.0$ )  | 0.313        | 0.754   | 0.287  |
| VAE ( $\beta=1.0$ )           | 0.340        | 0.778   | 0.310  |
| $\beta$ -VAE ( $\beta=8.0$ )  | 0.350        | 0.763   | 0.299  |
| AE (baseline)                 | 0.356        | 0.781   | 0.336  |
| DenosingAE                    | 0.361        | 0.792   | 0.366  |
| $\beta$ -VAE ( $\beta=1.0$ )  | 0.372        | 0.776   | 0.338  |
| $\beta$ -VAE ( $\beta=0.01$ ) | <b>0.407</b> | 0.783   | 0.365  |

1) *Score Inversion*: The central finding is **score inversion**: illicit transactions reconstruct with *lower* MSE than licit ones, so the anomaly score must be  $-MSE$ . Automated laundering produces stereotyped, templated patterns that are easier to reconstruct than diverse legitimate activity. Direction is auto-detected on validation by comparing  $AUC(y, MSE)$  vs.  $AUC(y, -MSE)$ .

2)  *$\beta$ -VAE Analysis*: At  $\beta=0.01$  (reconstruction-dominated),  $F1 = 0.407$ —the best unsupervised non-graph result. As  $\beta$  increases above 1, the KL term compresses licit and illicit representations together, weakening the anomaly signal. The F1 plateau at  $\approx 0.36$ – $0.41$  reflects the hard ceiling of unsupervised reconstruction-based detection on this dataset.

3) *EVT Thresholding*: EVT thresholding (Eq. 2) collapses on DOMINANT and LSTM-AE because score distributions are concentrated near zero with thin tails—the GPD assigns an

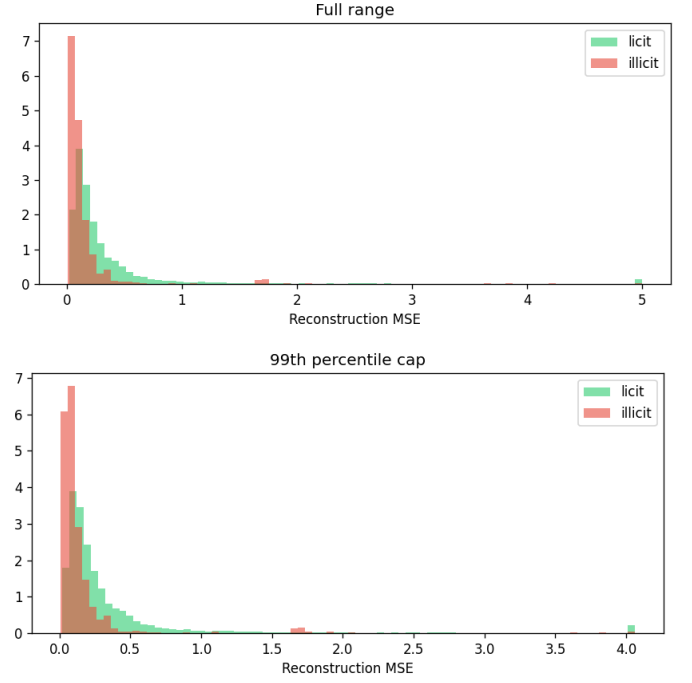


Fig. 3: Autoencoder reconstruction error distributions: histogram (top) and KDE (bottom) for licit (blue) and illicit (orange). Illicit transactions achieve lower MSE, confirming score inversion. Anomaly score is therefore  $-MSE$ .

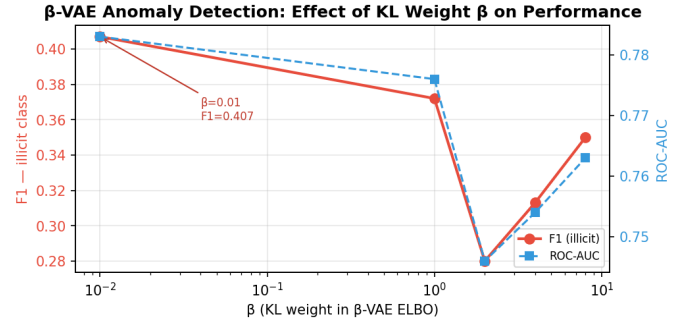


Fig. 4: Effect of KL weight  $\beta$  on  $\beta$ -VAE anomaly detection performance (F1 on illicit class and ROC-AUC, log- $\beta$  axis). Smaller  $\beta$  keeps the objective reconstruction-dominated, preserving the score-inversion signal. F1 peaks at  $\beta=0.01$  and degrades monotonically above  $\beta=1$  as the KL term conflates licit and illicit latent representations.

excessively high threshold, recalling almost no positives. EVT is effective only when reconstruction errors follow genuine heavy-tailed distributions.

4) *LSTM-AE*: The LSTM-AE aggregates features per time step ( $49 \times 165$ ), trains on 25 sliding windows ( $T=10$ ), and produces a time-step-level anomaly score. ROC-AUC = 0.446 ( $< 0.5$ , inverted) confirms the approach cannot discriminate individual transactions—illicit and licit rows within a step are averaged together. The most anomalous steps detected (40, 46, 39, 43, 36) correspond to high concept-drift periods, not

individual illicit transactions. Correct use: temporal monitoring, not per-transaction scoring.

#### D. DOMINANT: Graph Autoencoder

DOMINANT [6] jointly reconstructs node attributes and graph structure via a 2-layer GCN encoder (2 layers avoids over-smoothing on this sparse graph). Joint loss:

$$\mathcal{L} = \alpha \|\mathbf{X} - \hat{\mathbf{X}}\|_F^2 + (1 - \alpha) \text{BCE}(\mathbf{A}, \hat{\mathbf{A}}), \quad (4)$$

with  $\alpha=0.5$ . Result: **F1 = 0.212**, **AUC = 0.669**—the best fully unsupervised method, exceeding all classical detectors by  $\Delta > 0.13$ . An F1-optimal threshold is required (EVT collapses).

#### E. GNN Ablation Study

A GNN exploits *guilt by association*: money laundering chains link illicit wallets through intermediate transactions. Ten controlled experiments each isolate one design choice (Table VII).

TABLE VII: GNN Ablation — 10 Controlled Experiments

| Exp | Arch   | Configuration                       | F1           | AUC   | AP           |
|-----|--------|-------------------------------------|--------------|-------|--------------|
| 1   | SAGE   | 2L, h=64, 200 ep                    | 0.528        | 0.887 | 0.519        |
| 2   | SAGE   | 2L, h=128, 200 ep                   | 0.577        | 0.896 | 0.603        |
| 3   | SAGE   | 2L, h=256, 200 ep                   | 0.580        | 0.895 | 0.595        |
| 4   | SAGE   | 2L, h=128, 300 ep                   | 0.597        | 0.897 | 0.607        |
| 5   | SAGE   | 3L, h=128, 200 ep                   | 0.615        | 0.899 | 0.611        |
| 6   | SAGEv2 | 3L, h=256, LN+SL, 250 ep            | 0.683        | 0.906 | 0.704        |
| 7   | SAGE   | 3L, h=256, lr=0.0038, 274 ep (Opt.) | <b>0.690</b> | 0.901 | <b>0.706</b> |
| 8   | GAT    | 2L, 4 heads, lr=5e-4, 200 ep        | 0.307        | 0.824 | 0.366        |
| 9   | GAT    | 2L, 2 heads, lr=1e-3, 200 ep        | 0.341        | 0.864 | 0.419        |
| 10  | GATv2  | 3L, 2 heads, residual+LN, 300 ep    | 0.382        | 0.887 | 0.547        |

##### 1) Ablation Findings:

- **Width** (Exps 1–3, 64→256): +0.052 F1 total with diminishing returns; saturates beyond  $h=128$ .
- **Training duration** (Exps 2 vs. 4, +100 ep): +0.020 F1; cheapest lever before changing architecture.
- **Depth** (Exps 2 vs. 5, 2L→3L): +0.038 F1; 3-hop receptive field captures longer laundering chains.
- **LayerNorm + self-loops** (Exps 5 vs. 6): +0.068 F1, the largest single gain. LN prevents gradient instability at 3 layers; self-loops preserve node self-features during mean aggregation.
- **Optuna** (Exps 6 vs. 7, lr=0.0038): +0.007 F1; architecture choice dominates over tuning.
- **GAT family** (Exps 8–10): best GAT F1 = 0.382 vs. best SAGE F1 = 0.690 (gap = 0.308). On a sparse graph (avg. degree 2.3), attention weights over too few neighbors are noisy; mean aggregation is more robust.

**Answer to RQ3:** LayerNorm+self-loops > depth > width > tuning. In this configuration and feature setting, vanilla GAT and GATv2 are counterproductive on a sparse graph (avg. degree 2.3); recent work with Chebyshev convolutions and deeper GATv2 + skip connections has narrowed the gap on Elliptic under different feature subsets, so the finding should not be generalized beyond mean-aggregation architectures at this scale.



Fig. 5: GNN ablation across 10 experiments. Top: F1 progression showing the GraphSAGE improvement trajectory (Exps 1–7) and the GAT family ceiling (Exps 8–10). Bottom: corresponding AUC/AP values confirming the same ranking.

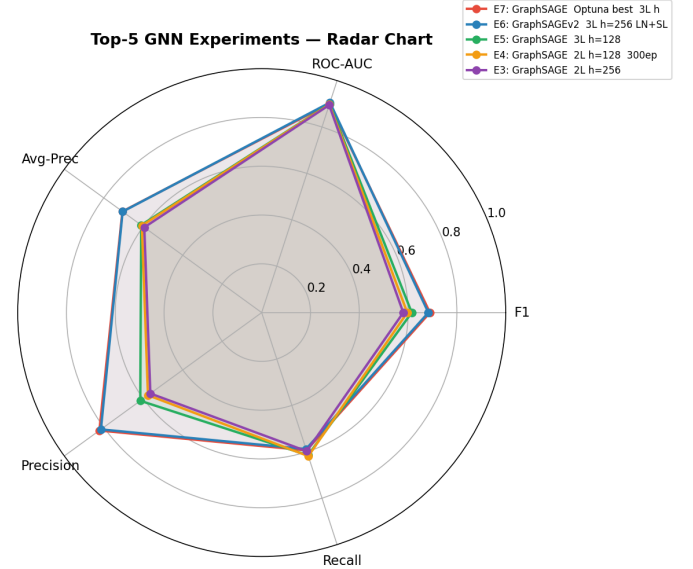


Fig. 6: Radar chart of the top-5 GNN experiments across five metrics: F1, ROC-AUC, Average Precision, Precision, and Recall. GraphSAGEv2 (Exp 6) and the Optuna-tuned variant (Exp 7) dominate across all axes, while GAT family experiments show systematically lower coverage, confirming mean aggregation outperforms attention on this sparse graph.

TABLE VIII: Optuna TPE Tuning Results (40 trials each)

| Model     | Base F1 | Tuned F1     | $\Delta$ |
|-----------|---------|--------------|----------|
| GBM       | 0.766   | <b>0.824</b> | +0.058   |
| GraphSAGE | 0.541   | 0.688        | +0.147   |
| RF        | 0.801   | 0.797        | −0.004   |

### F. Hyperparameter Tuning (Optuna)

Best parameters: GBM ( $n=245$ , depth= 8, lr= 0.121, sub= 0.918); GraphSAGE (hidden= 256, dropout= 0.45, lr= 0.0038, 274 ep); RF is near ceiling ( $-0.004$  within search variance). GBM profits from depth-lr interaction; GraphSAGE gains the most absolutely because default lr= 0.01 is too aggressive for this sparse graph.

### G. Ensemble and Calibration

TABLE IX: Ensemble and Calibration Results

| Model                        | F1    | ROC-AUC | PR-AUC |
|------------------------------|-------|---------|--------|
| Ensemble (RF+LGB+GAT)        | 0.742 | 0.922   | 0.789  |
| Ensemble (GBM+LGBM+SAGEv2)   | 0.821 | 0.924   | 0.802  |
| Weighted Ensemble (1/3 each) | 0.821 | 0.924   | 0.802  |
| GBM + Platt calibration      | 0.801 | 0.920   | 0.800  |
| GBM + Isotonic calibration   | 0.734 | 0.875   | 0.768  |

The GBM+LGBM+SAGEv2 ensemble (F1 = 0.821) narrows the gap to GBM solo (0.824) but does not surpass it. Testing four weight schemes shows equal weights win—downweighting SAGEv2 removes useful graph signal. GBM is already well-calibrated (Brier = 0.0196); Platt scaling makes probabilities more conservative ( $-0.023$  F1); isotonic regression overfits validation and distorts ordering (AUC drops 0.920 $\rightarrow$ 0.875).

### H. Full 48-Experiment Leaderboard

Table X reports the top 25 of 48 experiments. Fig. 7 shows the top-15 by F1. Fig. 8 shows Precision–Recall curves for key supervised models, confirming that high AP ( $\approx 0.80$ ) is achievable across the recall range.

TABLE X: Top-25 of 48 Experiments (by F1 on illicit class)

| #  | Model                         | Cat.     | F1           | AUC   | AP    |
|----|-------------------------------|----------|--------------|-------|-------|
| 1  | GBM + Structural              | Sup.     | <b>0.827</b> | 0.924 | 0.803 |
| 2  | GBM (Optuna)                  | Sup.     | 0.824        | 0.920 | —     |
| 3  | GBM + PseudoLabels            | Semi     | 0.824        | 0.924 | 0.804 |
| 4  | Ensemble GBM+LGBM+SAGEv2      | Ens.     | 0.821        | 0.924 | 0.802 |
| 5  | Ensemble equal 1/3            | Ens.     | 0.821        | 0.924 | 0.802 |
| 6  | GBM + Temporal Roll.          | Sup.     | 0.818        | 0.930 | 0.802 |
| 7  | LightGBM                      | Sup.     | 0.817        | 0.930 | 0.804 |
| 8  | RF + Graph Features           | Sup.     | 0.807        | 0.942 | 0.800 |
| 9  | RF (baseline)                 | Sup.     | 0.801        | 0.935 | 0.794 |
| 10 | GBM + Platt                   | Sup.     | 0.801        | 0.920 | 0.800 |
| 11 | RF (Optuna)                   | Sup.     | 0.797        | 0.929 | —     |
| 12 | GBM (baseline)                | Sup.     | 0.766        | 0.914 | 0.785 |
| 13 | Ensemble RF+LGB+GAT           | Ens.     | 0.742        | 0.922 | 0.789 |
| 14 | GBM + Isotonic                | Sup.     | 0.734        | 0.875 | 0.768 |
| 15 | SAGEv2 + PseudoLabels         | Semi-GNN | 0.729        | 0.890 | 0.752 |
| 16 | GraphSAGEv2                   | GNN      | 0.698        | 0.913 | 0.720 |
| 17 | GNN Exp 7 (Optuna)            | GNN      | 0.690        | 0.901 | 0.706 |
| 18 | GraphSAGE (Optuna)            | GNN      | 0.688        | —     | —     |
| 19 | GNN Exp 6 (3L LN+SL)          | GNN      | 0.683        | 0.906 | 0.704 |
| 20 | GNN Exp 5 (3L h=128)          | GNN      | 0.615        | 0.899 | 0.611 |
| 21 | GraphSAGE (baseline)          | GNN      | 0.541        | 0.888 | 0.549 |
| 22 | $\beta$ -VAE ( $\beta=0.01$ ) | Neural   | 0.407        | 0.783 | 0.365 |
| 23 | DenoisingAE                   | Neural   | 0.361        | 0.792 | 0.366 |
| 24 | DOMINANT                      | Graph-AE | 0.212        | 0.669 | 0.123 |
| 25 | LSTM-AE                       | Neural   | 0.122        | 0.446 | 0.064 |

**Five performance tiers:**

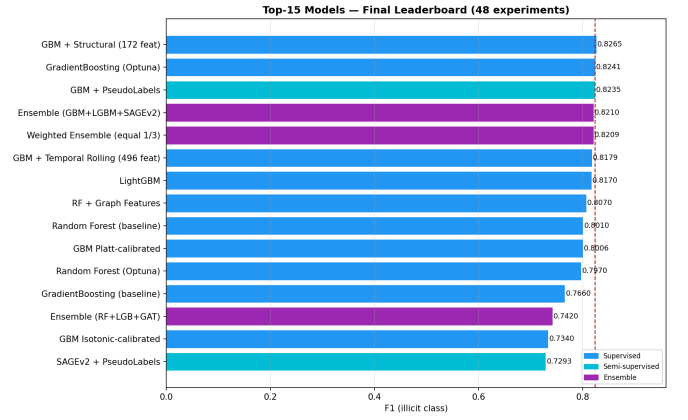


Fig. 7: Top-15 models by F1 on illicit class. The dashed line marks the GBM Optuna baseline (0.824) for reference. All top-15 entries are supervised or ensemble-based.

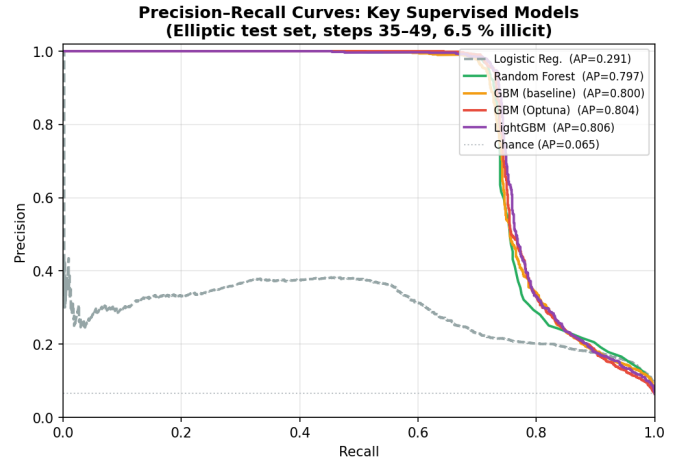


Fig. 8: Precision–Recall curves for key supervised models on the test set (steps 35–49, 6.5 % illicit). LightGBM and GBM (Optuna) achieve AP  $\approx 0.80$ , maintaining high precision across the recall range. Logistic Regression collapses to low precision at moderate recall (AP = 0.291). The dotted line marks the chance baseline (AP = 0.065). PR-AUC is threshold-independent and more informative than ROC-AUC under severe class imbalance.

- 1) **Supervised/Ensemble** (F1 0.74–0.83): tree models dominate due to 71 aggregated features pre-encoding graph signal.
- 2) **GNN supervised** (F1 0.54–0.73): GraphSAGEv2 recovers 84.5 % of the supervised ceiling using only topology.
- 3) **Neural AE** (F1 0.12–0.41): bounded by reconstruction ceiling; best from  $\beta$ -VAE ( $\beta=0.01$ ).
- 4) **Graph AE unsupervised** (DOMINANT, F1 = 0.212): best unsupervised *graph* method ( $\beta$ -VAE is higher overall).
- 5) **Classical unsupervised** (F1 < 0.10): density methods fail.



**Answer to RQ2:** graph structure alone recovers 84.5 % of the supervised ceiling. GBM and GraphSAGEv2 exploit different, complementary channels (tabular features vs. topology).

### I. Explainability

TABLE XI: Cross-Model Feature Agreement (top-20 lists)

| Agreement                    | Features                                           |
|------------------------------|----------------------------------------------------|
| RF $\cap$ GBM (7)            | 1f_18, 1f_47, 1f_53, 1f_59, 1f_76,<br>1f_90, af_70 |
| RF $\cap$ GBM $\cap$ GNN (3) | <b>1f_53, 1f_90, af_70</b>                         |

SHAP TreeExplainer was applied to GBM (Optuna, the best tabular model,  $F1 = 0.824$ ) and RF; gradient attribution was computed for GraphSAGE by backpropagating the illicit-class logit to the input and averaging over illicit test nodes. Fig. 9 shows the GBM beeswarm as the primary explainability view.

Three features—1f\_53, 1f\_90 (local), and af\_70 (aggregated neighborhood)—rank top-20 in every model family. Their cross-paradigm consensus validates them as genuine signals, not model-specific artifacts. The presence of the aggregated feature af\_70 in GNN attribution confirms that graph structure encodes real anomaly signal.

**Answer to RQ4:** robust cross-paradigm signals exist spanning both local transaction statistics and neighborhood aggregates. Identifying their semantics via Elliptic collaboration is the primary open question.

## V. DISCUSSION

**RQ1 — Density failure.** All classical detectors score below  $F1 = 0.076$ . Illicit transactions are stereotyped, not rare. This directly contradicts the assumption of classical anomaly detection—that anomalies are low-density outliers. On the Bitcoin blockchain, money-laundering patterns (chain peeling, mixing) are individually indistinguishable from legitimate high-frequency activity.

**RQ2 — Graph vs. tabular.** GraphSAGEv2 ( $F1 = 0.698$ ) recovers 84.5 % of GBM ( $F1 = 0.824$ ). The remaining 15.5 % gap reflects the richer signal encoded in the 71 pre-computed aggregated features available to tree models. A stacking ensemble of GBM + GraphSAGEv2 is the natural next step.

**RQ3 — GNN design.** LayerNorm+self-loops (+0.068) > depth (+0.038) > width (+0.052 total) > learning-rate tuning (+0.007). Mean aggregation dominates attention on sparse graphs.

**RQ4 — Robust features.** 1f\_53, 1f\_90, af\_70 are top-20 across all model families. The aggregated feature af\_70 in GNN attribution confirms that graph neighborhood signal is captured independently by tabular proxies in tree models.

**Autoencoder paradox.** Illicit transactions are *more* regular than licit ones. Score inversion (low MSE = anomaly) must be validated on labeled data before deploying any reconstruction-based AML system.

**$\beta$ -VAE.** Lower  $\beta$  preserves reconstruction dominance;  $\beta \geq 2$  compresses licit and illicit representations together. The ceiling

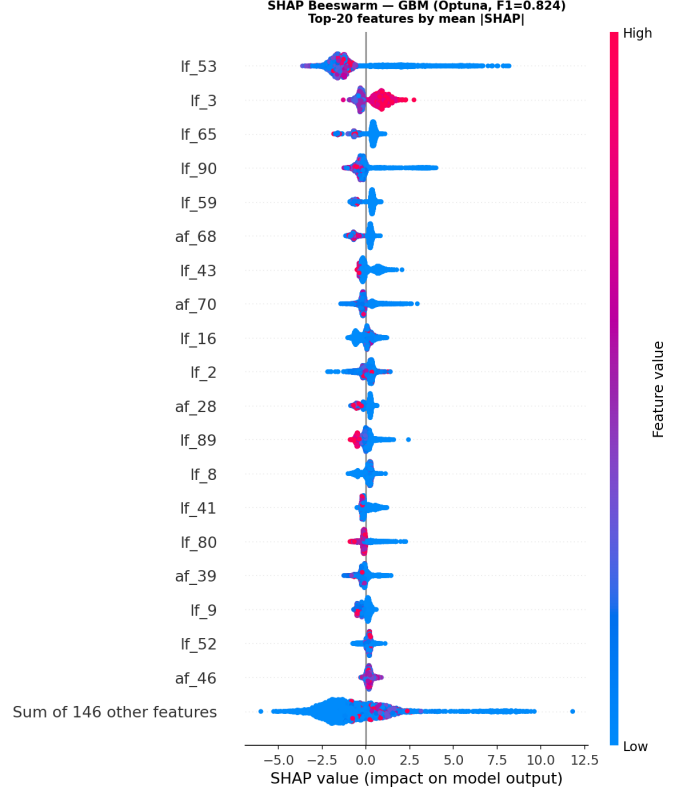


Fig. 9: SHAP beeswarm for GBM (Optuna,  $F1 = 0.824$ ), the best tabular model. Each dot is one test-set transaction (sample of 3 000); color encodes feature value (red=high, blue=low). Positive SHAP pushes toward illicit; negative toward licit. Features 1f\_53, 1f\_90, and af\_70 rank in the top band and appear consistently across RF, GBM, and GraphSAGE gradient attribution.

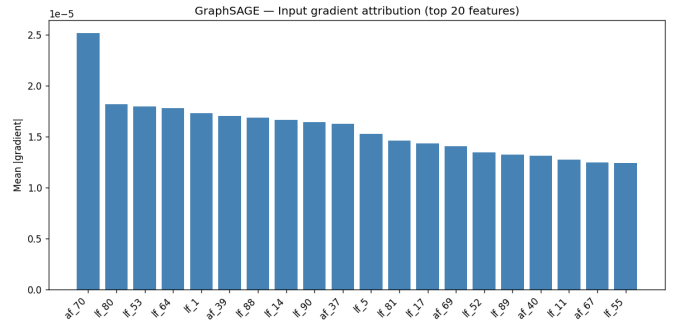


Fig. 10: GraphSAGE gradient attribution: mean absolute gradient per feature, averaged over illicit test nodes. Features 1f\_53, 1f\_90, and af\_70 are prominent despite the entirely different learning mechanism, confirming the cross-paradigm consensus.

at  $F1 \approx 0.41$  is the fundamental limit of unsupervised reconstruction-based detection here.

**Semi-supervised pseudo-labels.** GBM gains  $\Delta = -0.0006$ : supervised boundary is already saturated. GraphSAGEv2 gains  $+0.031$ : extra pseudo-labeled nodes populate graph neighborhoods during message passing.

**Concept drift.** Illicit rate drops from 11.6% to 6.5% over 15 steps. Production systems need temporal threshold recalibration; practical strategies include sliding-window F1 re-optimization on the most recent labeled window, Bayesian prior-shift correction, and adaptive class-weight schedules tied to rolling illicit-rate estimates.

**Limitations.** (1) Feature semantics unknown; robust features cannot be interpreted without Elliptic collaboration. (2) GAT Optuna tuning incomplete; recent Elliptic work with Chebyshev convolutions and skip-connected GATv2 reports higher GNN F1 under different feature subsets, so our GNN results represent a strong but not exhaustive upper bound. (3) GBM + GraphSAGEv2 stacking unexplored. (4) DOMINANT with more epochs and  $\alpha$  tuning may gain further. (5) Per-transaction LSTM-AE (vs. timestep aggregate) not evaluated. (6) This work positions itself as a comprehensive baseline benchmark across paradigms; direct comparison to reported Elliptic state-of-the-art (e.g., Chebyshev-GCN, N2V-GCN) is left for future work.

## VI. CONCLUSION

We benchmarked 48 model variants on the Elliptic dataset across five improvement rounds. Six actionable findings:

- 1) **Temporal split is mandatory.** Random shuffling leaks future patterns; the 11.6%→6.5% illicit-rate drift is real.
- 2) **Supervised trees dominate.** GBM + structural topology reaches  $F1 = 0.827$ ; 71 aggregated features encode enough graph signal to outperform unsupervised GNNs.
- 3) **GNNs add complementary structure.** GraphSAGEv2 reaches  $F1 = 0.698$  using only graph topology. 3-layer mean aggregation with LayerNorm and self-loops is optimal; GAT consistently underperforms on sparse graphs.
- 4) **Reconstruction detection is viable but inverted.**  $\beta$ -VAE ( $\beta=0.01$ ,  $F1 = 0.407$ ) is the best unsupervised non-graph method. Score direction must be validated before deployment.
- 5) **DOMINANT is the best unsupervised graph-autoencoder method.**  $F1 = 0.212$ ,  $AUC = 0.669$ , exceeding all classical density detectors by  $>0.13$   $F1$  ( $\beta$ -VAE, tabular, reaches  $F1 = 0.407$ ).
- 6) **Three features are cross-paradigm robust.** `lf_53`, `lf_90`, `af_70` rank top-20 in RF, GBM (SHAP), and GraphSAGE (gradient attribution).

**Future work:** (1) GBM + GraphSAGEv2 stacking ensemble; (2) per-transaction LSTM-AE; (3) EvolveGCN [2] for temporal drift; (4) online threshold recalibration; (5) GNNExplainer for node-level attribution; (6) Elliptic collaboration to identify feature semantics.

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